

DIPLOMATIC WORKING DOCUMENT — MODULAR ANNEX

ANNEX Y

Dual-Key Custody Protocol

For In-Country HEU Storage Under Joint IAEA / U.S.-Designated Control

MODULAR • INCORPORABLE BY REFERENCE • STEP-BACK / EARN-
BACK INTEGRATED

Version 1.0 — June 2026

AUTHORIZED NEGOTIATORS ONLY — CONFIDENTIAL — NOT FOR PUBLIC DISTRIBUTION

TABLE OF CONTENTS

Preamble

Purpose

Background

Relationship to Parent Framework

Modularity Clause

Article 1: The Dual-Key Store

1.1 Facility and Sealing

1.2 Key Holders

1.3 Operational Rules

1.4 Pre-Execution Escrow Verification

Article 2: Routine Access and Inspections

2.1 Scheduled Joint Inventory

2.2 Unscheduled Access

Article 3: Material Withdrawal

3.1 Permitted Withdrawals

3.2 Withdrawal Procedure

3.3 Unauthorized Withdrawal

Article 4: Custody Termination (After Day 365)

Article 5: Step-Back / Earn-Back Integration

Article 6: Relationship to Other Annexes

Article 7: Incorporation by Reference

Signature Block

PREAMBLE

Purpose

The Parties establish a Dual-Key Custody Protocol for any highly enriched uranium or other fissile material that Iran stores intact — rather than downblending or transforming — during Phase I or Phase II. The material resides in a single, sealed, IAEA-designated storage facility at Esfahan or another mutually agreed location. Two independent digital keys control access: one key resides with the IAEA; one key resides with a U.S.-designated technical agent. Neither Party opens the store alone.

Iran retains the sovereign right to keep enriched uranium material on Iranian soil. The United States requires that no weapons-usable material remains under Iran's sole control. This Annex reconciles both positions through a joint custody architecture that excludes unilateral access by either Party.

All obligations in this Annex execute in the active present tense. Breaches trigger the Three-Tier Step-Back system set out in Annex T.

Background

The parent framework — the Memorandum of Understanding Pertaining to PEACE — establishes dual-key custody as the default storage modality for intact HEU during Phase I and Phase II. The MOU Article 3.2 provides that in-country material storage operates under the Dual-Key Custody Protocol, with the IAEA holding one key and a U.S.-designated technical agent holding the second. The MOU Article 3.2(b) preserves Iran's right to request termination of IAEA custody and return to sovereign control after Day 365, subject to the conditions specified therein.

This Annex operationalizes those provisions. It specifies the facility, the key architecture, access procedures, material withdrawal rules, custody termination protocols, and the integration with the step-back / earn-back enforcement system. The Annex constitutes a self-contained, modular instrument that incorporates into the parent framework by reference without requiring amendment.

Relationship to Parent Framework

This Annex supplements and does not replace the custody architecture established in the parent framework. All obligations under Article 3 (Nuclear Stabilization Core) remain in full force. Where this Annex provides specific protocols for dual-key custody operations, the provisions of this Annex govern.

Modularity Clause

This Annex is self-contained and incorporates into any existing or future agreement between the Parties by a single sentence of adoption. It does not amend or supersede conflicting provisions of any parent framework unless the Parties explicitly so agree. In the event of inconsistency between this Annex and any parent framework provision, this Annex governs with respect to joint custody of intact HEU stored in-country.

ARTICLE 1: THE DUAL-KEY STORE

OPERATIVE AND BINDING

1.1 Facility and Sealing

Iran designates a single storage location — the Joint Custody Vault — at the Esfahan tunnel complex or at an alternative site the Parties mutually agree. The IAEA verifies the vault's structural integrity, seals all access points with tamper-evident, cryptographically signed seals, and installs continuous dual-camera monitoring with encrypted feeds to Vienna, Tehran, and the Joint Secretariat in Muscat.

Once sealed, the vault remains closed except for joint inspection or material withdrawal under Article 3 of this Annex.

1.2 Key Holders

The dual-key system operates through two independent key holders, each maintaining autonomous control over one digital key:

Key	Holder	Authorization Scope
IAEA Key	IAEA Director General or designated senior inspector	Authorizes any vault opening from the verification and safety standpoint
U.S.-Designated Key	Non-uniformed technical agent of the United States, approved by Iran on a per-visit basis (per MOU Article 3.5)	Provides the second factor of control; no operational authority over vault contents

Both keys are digital, time-synchronized, and require biometric or hardware-token authentication. Each key holder maintains a backup key in a separate geographically distributed secure facility. The IAEA backup key resides in Vienna. The U.S.-designated backup key resides in Zurich.

The U.S.-Designated Key and the IAEA Key generate physically inside a secure server hosted by the Central Bank of Oman in Muscat. Neither key exists in extractable form outside this hardware security module. The Central Bank of Oman maintains the physical infrastructure but holds no logical access to either key material. Both Parties audit the hardware security module configuration prior to MOU entry into force.

1.3 Operational Rules

No opening of the vault occurs without both keys presented within a 60-second window.

Any attempted unilateral opening — using only one key — triggers an automatic Tier 2 Material Disruption under Annex T, locks the vault for 72 hours, and notifies all Parties and guarantor states within 5 minutes.

The Shared Digital Registry (Annex U) logs every key activation, whether successful or failed.

Rule	Specification
Dual-key requirement	Both keys presented within a 60-second window for any vault opening
Unilateral opening attempt	Automatic Tier 2 Material Disruption; 72-hour vault lock; notification to all Parties and guarantor states within 5 minutes
Digital registry	The Shared Digital Registry (Annex U) logs every key activation, successful or failed
Backup keys	Geographically distributed: IAEA backup in Vienna; U.S. backup in Zurich
Authentication	Biometric or hardware-token authentication required for each key
Day Zero Synchronization	U.S. Executive Order waivers unfreezing the first \$2 Billion maritime tranche are cryptographically locked in the same registry. The release of the U.S. digital token to Iran's vault interface occurs automatically and instantaneously upon the automated timestamp verifying the U.S. signature deposit.

1.4 Pre-Execution Escrow Verification

The Day-Zero Cryptographic Escrow implements a simultaneous blind switch that resolves the "who goes first" dilemma. Before any physical material moves, both the U.S. sanctions release token and Iran's material access token are pre-positioned in the Shared Digital Registry but remain in a cryptographically locked state.

The unlocking mechanism operates as follows: the U.S. Executive Order waivers for the first \$2 Billion maritime tranche are encoded as a digital token and deposited in the escrow registry. Iran's corresponding material disposition token is encoded and deposited in the same registry. Both tokens remain locked until the automated timestamp verification confirms simultaneous release conditions.

The release triggers automatically when: (a) the IAEA confirms Iran's material is in escrow-ready status at the designated facility; (b) the U.S. Treasury confirms waiver tokens are deposited and cryptographically signed; and (c) the Central Bank of Oman server verifies both conditions through its hardware security module and generates the synchronized release timestamp. Neither Party can unilaterally trigger or block the release once all three conditions register as satisfied.

If either Party fails to deposit its token within the agreed pre-execution window, the escrow remains locked and no material transfer or sanctions relief proceeds. The Shared Digital Registry logs all deposit attempts, successful releases, and lock conditions. Either Party requests a technical review of the escrow status through the PCC, which convenes within 48 hours to resolve any discrepancies.

This Pre-Execution Escrow Verification applies exclusively to the first tranche. Subsequent tranches follow the sequencing schedule in Annex S, with escrow verification repeating for each milestone.

ARTICLE 2: ROUTINE ACCESS AND INSPECTIONS

OPERATIVE AND BINDING

2.1 Scheduled Joint Inventory

Every 30 days, the IAEA and the U.S.-designated agent conduct a joint inventory of the vault's contents. The IAEA performs non-destructive assay — gamma spectroscopy and neutron counting — on each container. Both key holders are present to open the vault. The inspection lasts no more than 8 hours.

The IAEA issues an Inventory Verification Certificate within 24 hours of the inspection's completion. A clean certificate triggers any scheduled economic tranche linked to custody milestones.

2.2 Unscheduled Access

Either Party requests an unscheduled inspection if telemetry or national technical means indicates an anomaly. The request routes through the PCC. The IAEA evaluates the request within 6 hours. If the IAEA deems the request reasonable, both key holders open the vault within 48 hours.

Refusal by either key holder to participate in an IAEA-approved unscheduled inspection constitutes a Tier 2 violation — or Tier 3 if accompanied by evidence of diversion.

ARTICLE 3: MATERIAL WITHDRAWAL

OPERATIVE AND BINDING

3.1 Permitted Withdrawals

Material removal from the Dual-Key Vault occurs only for the following purposes:

- [1] **Downblending under Pathway A (in-country):** The material moves directly to the Esfahan downblending facility under joint escort.
- [2] **Export under Pathway B (Oman):** Iran exercises its unilateral, revocable right; the material moves under IAEA chain-of-custody to Oman.
- [3] **Transformation under Pathways N, O, T, or W:** The material moves to the designated transformation facility under joint escort.

3.2 Withdrawal Procedure

Iran notifies the PCC and the IAEA of the intended withdrawal at least 72 hours in advance, specifying the pathway, quantity, and destination facility.

The IAEA verifies that the withdrawal is consistent with the selected pathway.

Both key holders open the vault simultaneously. The IAEA seals each removed container with a unique tamper-evident seal and logs the GPS-tracked movement.

The Joint Secretariat monitors the transport in real time.

3.3 Unauthorized Withdrawal

Any removal of material from the vault without the procedure set out in Article 3.2 — including a unilateral opening of the vault — constitutes a Tier 3 Deliberate Hostile Violation. The consequences include:

- Full escrow freeze for the nuclear module.
- Immediate referral to the Guarantor Matrix.
- UNSC notification within 24 hours.

ARTICLE 4: CUSTODY TERMINATION (AFTER DAY 365)

OPERATIVE AND BINDING

Per Article 3.2(b) of the main MOU, Iran retains the right to request termination of IAEA custody and return to sovereign control on 180 days' notice, provided no downblending or transformation has commenced.

Upon delivery of such notice, the following sequence activates:

Custody Termination Sequence

1. The vault remains sealed under dual-key for the 180-day notice period.
2. The IAEA continues monthly inspections throughout the notice period.
3. At the end of 180 days, both key holders open the vault for a final joint inventory. Iran then assumes sole control. The IAEA removes its key and monitoring equipment. The U.S.-designated agent removes its key.
4. Escrow funds already disbursed before the termination notice remain retained. Undisbursed tranches tied to the terminated custody arrangement suspend but do not forfeit; they reinstate if Iran resumes IAEA custody within 180 days.

ARTICLE 5: STEP-BACK / EARN-BACK INTEGRATION

OPERATIVE AND BINDING

Any violation of this Annex triggers the tiered system of Annex T. The following table specifies the relationship between violation type, tier classification, and consequence:

Violation	Tier	Typical Consequence
Administrative delay in key availability (under 48h)	Tier 1	No penalty; technical assistance provided
Refusal to participate in scheduled inspection	Tier 2	10–25% reduction in nuclear-linked benefits
Attempted unilateral vault opening (no material removed)	Tier 2	25% reduction; 30-day remediation window
Unauthorized material removal	Tier 3	50% reduction; full escrow freeze; Guarantor Matrix referral

Earn-back operates as follows: full restoration occurs upon IAEA certification that remediation is complete and 14 clean days have passed for Tier 2 violations, or 90 clean days have passed for Tier 3 violations.

ARTICLE 6: RELATIONSHIP TO OTHER ANNEXES

OPERATIVE AND BINDING

This Annex operates in conjunction with the following annexes of the parent framework. The relationship is supplementary: this Annex governs dual-key custody specifics; the referenced annexes govern the domains indicated.

Annex	Relationship
Annex O (IAEA Access)	This Annex supplements Annex O with dual-key specificity for intact material custody
Annex V (Nuclear Dust)	Annex V governs entombed material, not intact cylinders under this Annex
Annex T	Annex T governs all penalties and earn-back procedures referenced in this Annex
Annex U	The Shared Digital Registry logs all vault events under this Annex
Annex W	Annex W provides analogous dual-key architecture for maritime transit

ARTICLE 7: INCORPORATION BY REFERENCE

OPERATIVE AND BINDING

This Annex incorporates into any existing or future agreement between the Parties by the following adoption sentence:

"The Parties adopt Annex Y (Dual-Key Custody Protocol) as an operative provision of this Agreement. In the event of any inconsistency, Annex Y governs with respect to joint custody of intact HEU stored in-country."

No amendment of the parent framework is required. This Annex enters into force upon signature by both Parties or provisional application by exchange of diplomatic notes.

SIGNATURE BLOCK

In witness whereof, the undersigned, being duly authorized by their respective governments, have signed this Annex Y: Dual-Key Custody Protocol.

Done at _____, this _____ day of _____, 2026, in triplicate in the English language.

For the Islamic Republic of Iran:

Name: _____

Title: _____

Date: _____

For the United States of America:

Name: _____

Title: _____

Date: _____

For the IAEA (as key holder and verification co-lead):

Name: Rafael Mariano Grossi, Director General

Date: _____

Witnessed by the Joint Secretariat (Muscat).